

$$20.1) \eta'' = x + \cos x. \quad \eta' = \int \eta'' dx = \frac{1}{2}x^2 + \sin x + C_1 \quad \eta = \int \eta' dx = \frac{1}{6}x^3 - \cos x + C_1 x + C_2$$

$$13). \text{令 } u = \eta'. \text{ 有 } 4xu' - u' = 4u \Rightarrow (4x-1)u' = 4u \Rightarrow (4x-1) \frac{du}{dx} = 4u \Rightarrow \frac{1}{4u} du = \frac{1}{4x-1} dx$$

$$\text{两边积分: } \frac{1}{4} \ln|u| + C_1 = \frac{1}{4} \ln|4x-1| + C_2 \Rightarrow \ln|u| = \ln|4x-1| + C \Rightarrow u = C(4x-1) = \eta'$$

$$\therefore u = C_1(2x^2 - x) + C_2.$$

$$14). \text{令 } u = \eta'. \quad u' = \eta'' = \frac{d^2 \eta}{dx^2} = \frac{d}{dx} \left(\frac{d\eta}{dx} \right) = \frac{du}{dx} = \frac{du}{d\eta} \cdot \frac{d\eta}{dx} = \frac{du}{d\eta} \cdot u$$

$$\therefore \eta \cdot u \frac{du}{d\eta} - u^2 = 0$$

$$1^\circ \text{ 若 } u=0. \quad \eta = C$$

$$2^\circ \text{ 若 } u \neq 0. \text{ 有: } \frac{\eta du}{d\eta} = u \Rightarrow \frac{1}{\eta} d\eta = \frac{1}{u} du \Rightarrow \ln|\eta| = \ln|u| + C$$

$$\therefore u = C_1 \eta \Rightarrow \eta' = \frac{d\eta}{dx} = C_1 \eta \Rightarrow \frac{1}{\eta} d\eta = C_1 dx \Rightarrow C_1 x + C_2 = \ln|\eta|$$

$$\therefore \eta = e^{C_1 x + C_2} = C_3 e^{C_1 x} \quad 2u$$

$$15). \text{令 } u = \eta' \quad u' = \eta''. \text{ 有 } u' = u^2 + u \Rightarrow \frac{du}{dx} = u^2 + u \Rightarrow \frac{1}{u^2+u} du = dx \Rightarrow \left(\frac{1}{u} - \frac{u}{u^2+1} \right) du = dx$$

$$\text{两边积分: } \ln|u| - \frac{1}{2} \ln|u^2+1| + C_1 = x + C_2 \Rightarrow \frac{|u|}{\sqrt{u^2+1}} = C e^x \Rightarrow \frac{u^2}{u^2+1} = C^2 e^{2x}$$

$$\frac{1}{C^2 e^{2x}} = 1 + \frac{1}{u^2} \Rightarrow \frac{1 - C^2 e^{2x}}{C^2 e^{2x}} = \frac{1}{u^2} \Rightarrow u^2 = \frac{C^2 e^{2x}}{1 - C^2 e^{2x}} \Rightarrow u = \frac{C e^x}{\sqrt{1 - C^2 e^{2x}}}$$

$$\therefore \eta = \int \frac{C e^x}{\sqrt{1 - C^2 e^{2x}}} dx \quad \text{令 } u = C e^x. \quad du = C e^x \quad \therefore \eta = \int \frac{u}{\sqrt{1 - u^2}} dx = \arcsin(C e^x) + C_1$$

$$16). \text{令 } \eta' = u \quad u' = \eta'''. \text{ 有 } u = u' = \frac{du}{dx} \Rightarrow \frac{1}{u} du = dx \Rightarrow \ln|u| = x + C \Rightarrow u = C_1 e^x = \eta''$$

$$\therefore \eta' = \int C e^x dx = C_1 e^x + C_2 \quad \eta = \int C_1 e^x + C_2 dx = C_1 e^x + C_2 x + C_3$$

$$21.11) y'' = (y')^2. \text{ 令 } u = y' \Rightarrow u' = y''. \text{ 有 } u' = u^2 \Rightarrow \frac{du}{dx} = u^2 \Rightarrow u^{-2} du = dx. \text{ 两边积分.}$$

$$-u^{-1} = x + C \Rightarrow -\frac{1}{y'} = x + C. \text{ 有 } y'(0) = -1 \Rightarrow C = 1$$

$$\therefore -\frac{1}{y'} = x + 1 \Rightarrow y' = \frac{-1}{x+1} \Rightarrow \text{两边积分: } y = -\int \frac{1}{x+1} dx \Rightarrow y = -\ln|x+1| + C. \quad y(0) = 0$$

$$\Rightarrow C = 0.$$

$$\therefore y = -\ln|x+1|$$

$$(3) \text{ 令 } u = y' \quad y'' = u' = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{du}{dx} = \frac{du}{dy} \cdot \frac{dy}{dx} = \frac{du}{dy} \cdot u$$

$$\therefore y \cdot \frac{du}{dy} \cdot u = u^2 - u^3. \text{ 当 } u=0 \text{ 时, } y=C$$

$$\text{当 } u \neq 0 \text{ 时, 有: } y \frac{du}{dy} = u - u^3 \Rightarrow \frac{1}{y} dy = \frac{1}{u(1-u)} du = \left(\frac{1}{1-u} + \frac{1}{u} \right) du$$

$$\therefore \ln|y| = -\ln|1-u| + \ln|u| + C = \ln \left| \frac{u}{1-u} \right| + C. \quad y(1) = 1, \quad y'(1) = -1$$

$$\therefore y = C \left(\frac{u}{1-u} \right) \Rightarrow x=1 \Rightarrow 1 = C \left(\frac{1}{2} \right) \Rightarrow C = -2$$

$$\therefore y = \frac{-2u}{1-u} = \frac{2u}{u-1} \Rightarrow y \cdot u - y = 2u \Rightarrow (2-y)u = -y \Rightarrow u = \frac{y}{y-2} = y' = \frac{dy}{dx}$$

$$\therefore \left(\frac{y-2}{y} \right) dy = dx. \text{ 两边积分}$$

$$\int \left(1 - \frac{2}{y} \right) dy = \int dx \Rightarrow y - 2\ln|y| = x + C$$

$$1 = 1 + C, C = 0.$$

$$\therefore \text{有 } y - 2\ln(y) = x$$

$$\frac{y-x}{2} = \ln(y) \Rightarrow y = e^{\frac{y-x}{2}}$$

$$(4) \text{ 令 } y' = u. \quad u' = y''. \text{ 有: } xu' + xu^2 - u = 0 \Rightarrow xu' - u = -xu^2$$

$$\Rightarrow u' - \frac{u}{x} = -u^2. \text{ 令 } p = u^{-1} \quad p' = -\frac{1}{u^2} \cdot u'$$

$$\Rightarrow \frac{u'}{u^2} - \frac{1}{u} \cdot \frac{1}{x} = -1$$

$$\Rightarrow -P' - P \cdot \frac{1}{x} = -1 \Rightarrow P' + P \frac{1}{x} = 1$$

$$\therefore \text{有 } P = C(x) \cdot e^{-\int \frac{1}{x} dx} \quad C(x) = \int 1 \cdot e^{\int \frac{1}{x} dx} dx = \frac{1}{2}x^2 + C$$

$$\therefore P = (\frac{1}{2}x^2 + C) \frac{1}{x} = \frac{1}{2}x + \frac{C}{x}$$

$$\therefore \frac{1}{u} = \frac{1}{2}x + \frac{C}{x} = \frac{1}{y}, \text{ 当 } x=2 \text{ 有 } 1 = 1 + \frac{C}{2} \Rightarrow C=0.$$

$$\therefore \frac{1}{y} = \frac{1}{2}x \Rightarrow y' = \frac{2}{x} \Rightarrow y = 2 \ln|x| + C \text{ 当 } x=2, y=2 \therefore 2 = 2 \ln 2 + C$$

$$\therefore C = 2 - 2 \ln 2. \therefore y = 2 \ln x + 2 - 2 \ln 2$$

24. (1) 线性相关.

(2) 线性无关

(3) 线性无关

$$25. \because y_1 = e^x, y_1' = y_1'' = e^x$$

$$(1+2x-x^2)e^x + (x^2-3)e^x + 2(1-x)e^x = 0 \cdot e^x = 0. \therefore \text{无解}$$

$$y_2' = 2ax + b, y_2'' = 2a$$

$$\therefore (1+2x-x^2)2a + (x^2-3)(2ax+b) + 2(1-x)(ax^2+bx+c) = 0$$

$$\begin{cases} -2a+b+2a-2b = -b \\ 4a-6a+2b-2c = -2a+2b-2c \\ 2a-3b+2c = 0 \end{cases}$$

$$\therefore \begin{cases} b=0 \\ a=-c \end{cases}$$

$$\therefore y_2 = x^2 - 1 \quad \therefore \text{通解为: } y = C_1 e^x + C_2 (x^2 - 1)$$

$$26.11) y_1 = x. \quad x^2 y'' + x y' - y = 0 \Rightarrow y'' + \frac{1}{x} y' - \frac{1}{x^2} y = 0$$

$$\text{f.} \quad y_2 = y_1 \int \frac{1}{y_1^2} e^{-\int \frac{1}{x} dx} dx. \quad \int \frac{1}{x} dx = \ln|x|. \quad e^{-\int \frac{1}{x} dx} = \frac{1}{|x|}$$

$$y_2 = x \cdot \int \frac{1}{x^2} \cdot \frac{1}{|x|} dx = x \cdot \left(-\frac{1}{2} x^{-2} + C\right)$$

$$\therefore y_2 = \frac{1}{x}$$

$$\therefore y = C_1 x + C_2 \frac{1}{x}$$

$$12). \quad x^2 y'' - 2x y' + (x^2 + 2) y = 0 \Rightarrow y'' - \frac{2}{x} y' + \frac{x^2 + 2}{x^2} y = 0$$

$$y_1 = x \cos x. \quad \text{f.} \quad y_2 = x \cos x \int \frac{1}{x^2 \cos^2 x} e^{\int \frac{2}{x} dx} dx, \quad e^{\int \frac{2}{x} dx} = e^{2 \ln|x|} = e^{\ln x^2} = x^2$$

$$y_2 = x \cos x \int \frac{1}{x^2 \cos^2 x} \cdot x^2 dx = x \cos x \int \frac{1}{\cos^2 x} dx = x \cos x \int \sec^2 x dx$$

$$= x \cos x \tan x = x \sin x$$

$$\therefore y = C_1 x \cos x + C_2 x \sin x$$

$$27.11) y_1 = \ln x \quad y_1' = \frac{1}{x} \quad y_1'' = -\frac{1}{x^2}. \quad \text{f.} \quad \lambda: 1 - \ln x - 1 + \ln x = 0. \quad \text{f.} \quad \lambda$$

$$x^2 (\ln x - 1) y'' - x y' + y = 0 \Rightarrow y'' - \frac{1}{x(\ln x - 1)} y' + \frac{y}{x^2(\ln x - 1)} = 0$$

$$y_2 = y_1 \int \frac{1}{y_1^2} e^{\int \frac{1}{x(\ln x - 1)} dx} dx$$

$$\int \frac{1}{x(\ln x - 1)} dx. \quad \text{f.} \quad \ln x - 1 = u. \quad u' = \frac{1}{x}. \quad du = \frac{1}{x} dx$$

$$\therefore \int \frac{1}{x(\ln x - 1)} dx = \int \frac{du}{u} = \ln|u| = \ln(\ln x - 1)$$

$$\therefore e^{\ln(\ln x - 1)} = \ln x - 1$$

$$\therefore y_2 = \ln x \cdot \int \frac{\ln x - 1}{(\ln x)^2} dx. \quad \text{f.} \quad t = \ln x \Rightarrow x = e^t \quad dx = e^t dt$$

$$\therefore \int \frac{\ln x - 1}{(\ln x)^2} dx = \int \frac{t-1}{t^2} e^t dt = \int e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dt = \frac{e^t}{t} + C$$

$$\therefore y_2 = \ln x \cdot \frac{x}{\ln x} = x$$

$$\therefore y = C_1 \ln x + C_2 x$$

$$2) y_1 = e^x \quad (x-1)e^x - (x+1)e^x + 2e^x = xe^x - e^x - xe^x - e^x + 2e^x = 0$$

$$(x-1)y'' - (x+1)y' + 2y = 0 \Rightarrow y'' - \frac{x+1}{x-1}y' + \frac{2}{x-1}y = 0$$

$$y_2 = y_1 \int \frac{1}{y_1^2} e^{\int \frac{x+1}{x-1} dx} dx \quad \int \frac{x+1}{x-1} dx = \int 1 + \frac{2}{x-1} dx = x + 2\ln|x-1|$$

$$\therefore y_2 = e^x \int \frac{1}{e^{2x}} e^x \cdot (x-1)^2 dx = e^x \int \frac{(x-1)^2}{e^x} dx$$

$$\int \frac{(x-1)^2}{e^x} dx \quad \text{令 } u = (x-1)^2 \quad dv = e^{-x} \quad v = -e^{-x}$$

$$uv = u'v + uv' \quad \int uv' dx = uv - \int u'v dx \Rightarrow \int \frac{(x-1)^2}{e^x} dx = -e^{-x}(x-1)^2 + \int (2x-2)e^{-x} dx$$

$$\int (2x-2)e^{-x} dx = 2 \int (x-1)e^{-x} dx = -2xe^{-x}$$

$$\therefore \int \frac{(x-1)^2}{e^x} dx = -e^{-x}(x^2 - 2x + 1) - 2xe^{-x} = -e^{-x}(x^2 - 2x + 1 + 2x) = -(x^2 + 1)e^{-x}$$

$$\therefore y_2 = e^x \cdot (-x^2 - 1)e^{-x} = -x^2 - 1 \quad \text{取 } y_2 = x^2 + 1$$

$$\therefore y = C_1 e^x + C_2 (x^2 + 1)$$

28. $y^* = x$. 代入 $0 - 3x + 4x = x$. 成立. y^* 是原非齐次方程之解

$$y_1 = x^2 \quad y_2 = x^2 / \ln x \quad \text{代入: } y_1' = 2x \quad y_1'' = 2 \quad y_2' = 2x \ln x + x \quad y_2'' = 2 \ln x + 3$$

$$2x^2 - 6x^2 + 4x^2 = 0 \quad \text{成立}$$

$$2x^2 / \ln x + 3x^2 - 6x^2 / \ln x - 3x^2 + 4x^2 / \ln x = 0 \quad \text{成立}$$

$\therefore$ 是通解

$$29. (1) r^2 - 3r + 2 = 0 \quad r_1 = 2, \quad r_2 = 1$$

$$\therefore y = C_1 e^{2x} + C_2 e^x$$

$$(2) r^2 - 2r - 2 = 0. \quad r^2 - 2r + 1 = 3 \Rightarrow (r-1)^2 = 3 \quad r = 1 \pm \sqrt{3}$$

$$\therefore y = C_1 e^{(1+\sqrt{3})x} + C_2 e^{(1-\sqrt{3})x}$$

$$(3) r^2 - 2r = 0 \Rightarrow r_1 = 0, \quad r_2 = 2$$

$$\therefore y = C_1 + C_2 e^{2x}$$

$$(4) r^2 + 6r + 9 = 0. \quad (r+3)^2 = 0$$

$$\therefore r = -3$$

$$\therefore y = C_1 e^{-3x} + C_2 x e^{-3x}$$

$$(5) r^2 + 6r + 10 = 0. \quad r^2 + 6r + 9 = -1$$

$$\therefore (r+3)^2 = -1 \Rightarrow r+3 = \pm i$$

$$\therefore r = -3 \pm i$$

$$\therefore y = C_1 e^{-3x} \cos x + C_2 e^{-3x} \sin x$$

$$30. (1) r^2 - 2r + 2 = 0 \Rightarrow (r-1)^2 = -1 \quad \therefore r = 1 \pm i$$

$$y = C_1 e^x \cos x + C_2 e^x \sin x. \quad y(0) = C_1 = 1$$

$$y' = C_1 e^x (\cos x - \sin x) + C_2 e^x (\sin x + \cos x)$$

$$y'(0) = C_1 + C_2 = 2 \quad \therefore C_2 = 1 \quad \therefore \text{有 } y = e^x \cos x + e^x \sin x$$

$$(4). y^2 + 9 = 0 \Rightarrow y = \pm 3i$$

$$\therefore y = C_1 \cos 3x + C_2 \sin 3x$$

$$y\left(\frac{\pi}{3}\right) = 0 = C_1 \cos \pi + C_2 \sin \pi \Rightarrow C_1 = 0$$

$$y' = 3C_2 \cos 3x \Rightarrow y'\left(\frac{\pi}{3}\right) = 3C_2 \cos \pi = 1 \Rightarrow C_2 = -\frac{1}{3}$$

$$\therefore y = -\frac{1}{3} \sin 3x$$

$$31.11). r^2 - 3r = 0 \Rightarrow r_1 = 3, r_2 = 0$$

$$\therefore \text{齐次通解: } y = C_1 e^{3x} + C_2$$

$$x-1 = (1 \cdot x' + 1 \cdot x'') e^{0 \cdot x}$$

$$\lambda = 0 \text{ 单特征根}$$

$$\therefore y^* = x (B_0 x + B_1) = B_0 x^2 + B_1 x$$

$$y^{*'} = 2B_0 x + B_1, \quad y^{*''} = 2B_0$$

$$\therefore 2B_0 - 6B_0 x - 3B_1 = x - 1$$

$$\therefore B_0 = -\frac{1}{6}, \quad B_1 = \frac{2}{9}$$

$$\therefore y = C_1 e^{3x} + C_2 - \frac{1}{6} x^2 + \frac{2}{9} x$$

$$(2). 6r^2 - r - 2 = 0 \quad (3r-2)(2r+1) = 0. \quad r_1 = \frac{2}{3}, \quad r_2 = -\frac{1}{2}$$

$$\therefore \text{齐次通解: } y = C_1 e^{\frac{2}{3}x} + C_2 e^{-\frac{1}{2}x}$$

$$x e^x = (1 \cdot x + 0 \cdot x'') e^{1 \cdot x}$$

$$\therefore \lambda = 1 \quad k = 0$$

$$\therefore y^* = x^0 (Ax + B) e^x = (Ax + B) e^x$$

$$y^{*'} = (Ax + B + A) e^x$$

$$y^{*''} = (Ax + B + 2A) e^x$$

$$6(Ax + 2A + B) e^x - (Ax + A + B) e^x - 2(Ax + B) e^x = x e^x$$

$$\therefore \underline{6Ax} + \underline{12A} + \underline{6B} - \underline{Ax} - \underline{A} - \underline{B} - \underline{2Ax} - \underline{2B} = x$$

$$3Ax + 11A + 3B = x$$

$$\therefore \begin{cases} 11A + 3B = 0 \\ A = \frac{1}{3} \end{cases} \quad \therefore \begin{cases} A = \frac{1}{3} \\ B = -\frac{11}{9} \end{cases}$$

$$\therefore \text{通解: } y = C_1 e^{\frac{2}{3}x} + C_2 e^{-\frac{1}{2}x} + \left(\frac{1}{3}x - \frac{11}{9}\right) e^x$$

$$(3) x'' + 4x = t \sin t$$

$$r^2 + 4 = 0 \Rightarrow r = \pm 2i$$

$$\text{齐次通解: } x = C_1 \cos 2t + C_2 \sin 2t$$

$$t \sin t = [t \cdot \sin t + 0 \cdot \cos t] e^{0x}$$

$$\therefore \alpha = 0 \quad 0 + i \neq r \quad \therefore \text{非特征根}$$

$$\therefore x^* = (At + B) \sin t + (Ct + D) \cos t$$

$$x^{*'} = [(Ct + D) + A] \cos t + [C - (At + B)] \sin t$$

$$x^{*''} = [2C - At - B] \cos t + [-2A - Ct - D] \sin t$$

$$\therefore A = 0, C = \frac{1}{3}, B = -\frac{2}{9}, D = 0$$

$$\therefore x^* = -\frac{2}{9} \cos t + \frac{1}{3} t \sin t$$

$$\therefore \text{通解: } X = C_1 \cos 2t + C_2 \sin 2t - \frac{2}{9} \cos t + \frac{1}{3} t \sin t$$

$$(5) r^2 - 2r + 1 = 0 \Rightarrow r = 1$$

$$\therefore \text{齐次通解: } y = C_1 e^x + C_2 x e^x$$

令 2 个特解 y_1^* 和 y_2^*

y_1^* 对 $\frac{1}{2} e^x$, y_2^* 对 $\frac{1}{2} 3$

$$e^x = (0 \cdot x^0) e^x, \lambda = 1, \therefore k = 2$$

$$\therefore y_1^* = x^2 \cdot A \cdot e^x = A x^2 e^x$$

$$y_1^{*'} = 2A x e^x + A x^2 e^x$$

$$y_1^{*''} = 2A e^x + 2A x e^x + 2A x e^x + A x^2 e^x = 2A e^x + 4A x e^x + A x^2 e^x$$

$$\therefore 2A e^x + 4A x e^x + A x^2 e^x - 4A x e^x - 2A x e^x + A x^2 e^x = e^2$$

$$\therefore A = \frac{1}{2}$$

$$3 = e^0 (3 \cdot x^0), \lambda = 0, k = 0$$

$$\therefore y_2^* = B, y_2^{*'} = y_2^{*''} = 0$$

$$\therefore B = 3$$

$$\therefore y^* = \frac{1}{2} x^2 e^x + 3$$

$$\therefore \text{通解: } y = C_1 e^x + C_2 x e^x + \frac{1}{2} x^2 e^x + 3$$

$$32.11) r^2 - 3r + 2 = 0 \therefore r_1 = 2, r_2 = 1$$

$$\therefore \text{齐次通解: } y = C_1 e^{2x} + C_2 e^x$$

$$5 = x^0 (5x^0) e^{0x} \quad \lambda = 0 \quad k = 0$$

$$\therefore y^* = A. \quad y'' = y' = 0 \quad \therefore 2A = 5 \quad A = \frac{5}{2}$$

$$\therefore y^* = \frac{5}{2}$$

$$\therefore y = C_1 e^{2x} + C_2 e^x + \frac{5}{2}$$

$$y' = 2C_1 e^{2x} + C_2 e^x$$

$$C_1 + C_2 + \frac{5}{2} = 1 \Rightarrow C_1 + C_2 = -\frac{5}{2}$$

$$2C_1 + C_2 = 2 \quad \therefore C_1 = \frac{7}{2} \quad C_2 = -5$$

$$\therefore y = \frac{7}{2} e^{2x} - 5e^x + \frac{5}{2}$$

$$13). r^2 + 1 = 0 \Rightarrow r = \pm i$$

$$\therefore \text{齐次通解: } y = C_1 \cos x + C_2 \sin x$$

$$2 \cos x = 2 \cdot \cos x \cdot e^{0x} \quad \alpha = 0 \quad \beta = 1$$

$$0 \pm 1i = \pm i \quad c, k = 1$$

$$\therefore \text{设 } y^* = x (A \sin x + B \cos x). \quad y^{*'} = A \sin x + B \cos x + x(A \cos x - B \sin x)$$

$$y^{*''} = A \cos x - B \sin x + A \cos x - B \sin x + x(-A \sin x - B \cos x)$$

$$2A \cos x - 2B \sin x = 2 \cos x \quad A = 1 \quad B = 0.$$

$$\therefore y^* = x \sin x$$

$$\text{通解: } y = C_1 \cos x + C_2 \sin x + x \sin x. \quad y' = -C_1 \sin x + C_2 \cos x + \sin x + x \cos x$$

$$C_1 = 1. \quad C_2 = 0.$$

$$\therefore \text{特解: } y = \cos x + x \sin x$$

$$33. ar^2 + br + c = 0. \quad r = \frac{-b \pm \sqrt{\Delta}}{2a} \quad \Delta = b^2 - 4ac. \quad r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

1° 若 $b^2 - 4ac > 0$ $r_1 < 0$, $r_2 < 0$. 通解: $y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$ 在 $x \rightarrow \infty$ 时 $y \rightarrow 0$

2° 若 $b^2 - 4ac = 0$. $r = \frac{-b}{2a} < 0$ $y = C_1 e^{rx} + C_2 x e^{rx}$. 在 $x \rightarrow \infty$ 时 $y \rightarrow 0$

3° 若 $b^2 - 4ac < 0$. $r = -\frac{b}{2a} \pm i \frac{\sqrt{4ac - b^2}}{2a} = \alpha \pm \beta i$ $\alpha < 0$, $\beta > 0$

通解: $y = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$ $x \rightarrow \infty$ 时 $e^{\alpha x} \rightarrow 0$. 且 $C_1 \cos \beta x + C_2 \sin \beta x$ 有界

$\therefore y \rightarrow 0$

$\therefore$ 得证

$$34. f(x) = e^x + \int_0^x (t-x)f(t) dt = e^x + \int_0^x t f(t) dt - x \int_0^x f(t) dt$$

$$\Rightarrow f'(x) = e^x + x f(x) - \int_0^x f(t) dt - x f(x)$$

$$\therefore f'(x) = e^x - \int_0^x f(t) dt$$

$$f''(x) = e^x - f(x)$$

$$\therefore f''(x) + f(x) = e^x$$

$$\therefore r^2 + 1 = 0 \quad r = \pm i$$

$$\text{齐次通解: } y = C_1 \cos x + C_2 \sin x$$

$$e^x = (1 \cdot x^0) e^{1 \cdot x}$$

$\therefore \lambda = 1$ 非特征根.

$$k = 0$$

$$y^* = x^0 \cdot (B_0) e^x = B_0 e^x$$

$$y'' = B_0 e^x \quad 2B_0 e^x = e^x$$

$$\therefore B_0 = \frac{1}{2}$$

$$\therefore y^* = \frac{1}{2} e^x$$

$$\therefore \text{通解: } y = C_1 \cos x + C_2 \sin x + \frac{1}{2} e^x$$

$$f(0) = 1 \quad C_1 + \frac{1}{2} = 1 \Rightarrow C_1 = \frac{1}{2}$$

$$f'(0) = 1$$

$$y' = -C_1 \sin x + C_2 \cos x + \frac{1}{2} e^x$$

$$\therefore C_2 + \frac{1}{2} = 0 \Rightarrow C_2 = -\frac{1}{2}$$

$$\therefore y = \frac{1}{2} (\cos x + \sin x + e^x)$$

$$37. G = mg. \text{ 设 } F_{\text{阻}} = kv. \quad F_{\text{合}} = G - F_{\text{阻}} = mg - kv$$

$$\therefore a = \frac{dv}{dt} = g - \frac{kv}{m}$$

$$\therefore \frac{dv}{dt} + \frac{k}{m}v = g$$

$$\therefore v' + \frac{k}{m}v = g$$

$$\therefore v = C(x) e^{-\int \frac{k}{m} dt}. \quad C(x) = \int g e^{\int \frac{k}{m} dt} dt$$

$$\int \frac{k}{m} dt = \frac{k}{m} t$$

$$\therefore C(x) = g \int e^{\frac{k}{m} t} dt = \frac{m}{k} g e^{\frac{k}{m} t} + C_1$$

$$\therefore v = \left(\frac{mg}{k} e^{\frac{k}{m} t} + C_1 \right) \cdot e^{-\frac{k}{m} t}$$

$$= \frac{mg}{k} + C_1 e^{-\frac{k}{m} t}$$

$$V(0) = 0 \Rightarrow \frac{mg}{k} + C_1 = 0$$

$$V'(0) = g. \quad V' = -\frac{k}{m} C_1 e^{-\frac{k}{m}t}$$

$$\Rightarrow -\frac{k}{m} C_1 = g$$

$$C_1 = -\frac{mg}{k}$$

$$\therefore V = \frac{mg}{k} (1 - e^{-\frac{k}{m}t})$$

$$x = \int (V) dt = \frac{mg}{k} t + \frac{m^2 g}{k^2} e^{-\frac{k}{m}t} + C$$

$$x_0 = \frac{m^2 g}{k^2} + C = 0$$

$$\therefore C = -\frac{m^2 g}{k^2}$$

$$\therefore x(t) = \frac{mg}{k} t + \frac{m^2 g}{k^2} (e^{-\frac{k}{m}t} - 1)$$

$$39. 1) \frac{d^2 y}{dt^2} - 4 \frac{dy}{dt} + 3y = 0$$

$$\Delta r^2 - 4r + 3 = 0. \quad r_1 = 3. \quad r_2 = 1$$

$$\therefore \text{通解: } y = C_1 e^{3t} + C_2 e^t$$

$$t = \ln x$$

$$\therefore y = C_1 e^{3 \ln x} + C_2 e^{\ln x} = C_1 x^3 + C_2 x$$

$$(4) \text{ 令 } x = e^t. \text{ 有: } \frac{d^2 y}{dt^2} - 6 \frac{dy}{dt} + 8y = 2e^{3t}$$

$$r^2 - 6r + 8 = 0 \quad r_1 = 2. \quad r_2 = 4$$

$$\therefore \text{齐次通解: } y = C_1 e^{2t} + C_2 e^{4t}$$

$$2e^{3t} = (2 \cdot x^3) e^{3t}. \quad \lambda = 3 \quad k \neq 0$$

$$\therefore y^* = \chi^0(B) e^{3t}$$

$$y^* = 3Be^{3t} \quad y^{*''} = 9Be^{3t}$$

$$9Be^{3t} - 18Be^{3t} + 8Be^{3t} = 2e^{3t}$$

$$\therefore B = -2$$

$$\therefore y^* = -2e^{3t}$$

$$\therefore \text{通解: } y = C_1 x^3 + C_2 x^4 - 2x^3$$

$$b. \because y' + ay = f(x)$$

$$\therefore \text{有: } y = C(x) e^{-\int a dx} = C(x) e^{-ax}$$

$$C(x) = \int f(x) e^{\int a dx} dx = \int f(x) e^{ax} dx = \int_0^x f(t) e^{at} dt + C$$

$$\because \lim_{x \rightarrow \infty} f(x) = b. \therefore \exists X \text{ s.t. } x > X. \text{ 对 } \forall \varepsilon. \text{ 有 } |f(x) - b| < \varepsilon$$

$$\therefore y = e^{-ax} \left(\int_0^x f(t) e^{at} dt + C \right) = e^{-ax} \left(\int_0^x f(t) e^{at} dt + \int_x^\infty f(t) e^{at} dt + C \right)$$

$$= 0 + e^{-ax} \int_x^\infty f(t) e^{at} dt = b \int_x^\infty e^{at} dt = \frac{b}{a} (e^{ax} - e^{ax}) < \varepsilon \cdot \frac{e^{ax} - e^{ax}}{a}$$

$$\therefore \text{当 } x \rightarrow +\infty. \text{ 有 } y \rightarrow \frac{b}{a}$$

$$\text{当 } a < 0. \quad y(x) = e^{-ax} \left(\int_0^x f(t) e^{at} dt + C \right)$$

$$\lim_{x \rightarrow \infty} y(x) = \frac{b}{a}. \text{ 易知, } \frac{b}{a} \text{ 是一个有限常数.}$$

$$\therefore \lim_{x \rightarrow \infty} \left(\int_0^x f(t) e^{at} dt + C \right) = 0.$$

$$\therefore C = -\int_0^{+\infty} f(t) e^{at} dt. \therefore \text{只有一个解}$$

10. 污染物含量: $\frac{V}{b} \times \frac{Q_0}{V} = \frac{Q_0}{b}$ / 年.

设污染物量为 $Q(t)$. 流出量: $\frac{V}{3} \times \frac{Q(t)}{V} = \frac{Q(t)}{3}$

$$\therefore \frac{dQ}{dt} = \frac{Q_0}{b} - \frac{Q(t)}{3} = \frac{Q_0}{b} - \frac{Q}{3}$$

$$Q' + \frac{1}{3}Q = \frac{Q_0}{b}$$

$$Q = C(t) e^{-\int \frac{1}{3} dt}$$

$$C(t) = \int \frac{Q_0}{b} \cdot e^{\int \frac{1}{3} dt} dt = \frac{1}{b} Q_0 \int e^{\frac{1}{3}t} dt$$

$$Q = \left(\frac{1}{2} Q_0 e^{\frac{t}{3}} + C \right) e^{-\frac{t}{3}} = \frac{1}{2} Q_0 + C e^{-\frac{t}{3}}$$

$$\therefore Q(t) = \frac{1}{2} Q_0 + \frac{9}{2} Q_0 e^{-\frac{t}{3}}$$

$$\frac{Q_0}{2} + \frac{9Q_0}{2} e^{-\frac{t}{3}} \leq Q_0$$

$$-\frac{t}{3} \leq -\ln 9 \Rightarrow t \geq 6 \ln 3$$

(1.1.1) $\frac{dB}{dt} = 0.05B - 12000$

(2) $B' - 0.05B = -12000$

$$B = C(t) e^{\int 0.05 dt} = C(t) e^{0.05t}$$

$$C(t) = \int -12000 e^{-\int 0.05 dt} dt = -\int 12000 e^{-0.05t} dt = \frac{-12000}{-0.05} e^{-0.05t} + C$$

$$B = \left(\frac{12000}{0.05} e^{-0.05t} + C \right) e^{0.05t} = 240000 + C e^{0.05t}$$

$$B(0) = B_0$$

$$\therefore 240000 + C = B_0 \Rightarrow B(t) = 240000 + (B_0 - 240000) e^{0.05t}$$

(3) $B(20) = 0$.

$$\therefore 0 = 240000 + Ce \Rightarrow C = -\frac{240000}{e}$$

$$B_0 = 240000 - \frac{240000}{e} \approx 151704 \text{ (元)}$$

$$13. (x+1)f'(x) + (x+1)f(x) - \int_0^x f(t)dt = 0$$

$$\text{求导: } f'(x) + (x+1)f''(x) + \cancel{f(x)} + (x+1)f'(x) - \cancel{f(x)} = 0$$

$$\therefore (x+2)f'(x) + (x+1)f''(x) = 0$$

$$\text{令 } u = f'(x), u' = f''(x)$$

$$\therefore (x+2)u + (x+1)u' = 0$$

$$\therefore (x+2)u = -(x+1)u' = -(x+1) \frac{du}{dx}$$

$$\therefore \frac{x+2}{x+1} dx = -\frac{1}{u} du, \text{ 两边积分:}$$

$$(1 + \frac{1}{x+1}) dx = -\frac{1}{u} du$$

$$\therefore x + \ln|x+1| + C_1 = -\ln|u| + C_2$$

$$\therefore \ln|u| = -x - \ln|x+1| + C_3$$

$$u = e^{-x - \ln|x+1| + C_3} = C \frac{e^{-x}}{x+1} = y'$$

$$\therefore f'(x) = C \frac{e^{-x}}{x+1}$$

$$f(0) = 1, f'(0) = -1$$

$$\therefore C \frac{e^0}{1} = C = -1$$

$$\therefore f'(x) = \frac{-e^{-x}}{x+1}$$

$$(2) f'(x) = \frac{-e^{-x}}{x+1} < 0 \text{ (当 } x > 0 \text{ 时成立)}$$

$\therefore f(x)$ 在 $(0,1)$ 上 $\downarrow$.

$\therefore$ 当 $x > 0$, $f(x) < f(0) = 1$.

$$\text{令 } g(x) = f(x) - e^{-x}$$

$$g'(x) = f'(x) + e^{-x} > 0$$

$$\therefore g(x) > g(0) = 1 - 1 = 0$$

$$\therefore f(x) > e^{-x}$$

$\therefore$ 得证

$$15. (1) \text{ 特征方程: } r^3 - r^2 + 4r - 4 = 0$$

$$r^2(r-1) + 4(r-1) = (r^2+4)(r-1) = 0$$

$$r_1 = 1 \quad r_2 = 2i \quad r_3 = -2i$$

$$\therefore y = C_1 e^x + C_2 \sin 2x + C_3 \cos 2x$$

$$(2) \sin x \cdot \sin 2x = \frac{1}{2} \cos x - \frac{1}{2} \cos 3x$$

$$r^2 + 1 = 0 \quad r = \pm i$$

$$\text{齐次通解: } y = C_1 \cos x + C_2 \sin x$$

$$\text{设 } y^* = x(A \cos x + B \sin x) \Rightarrow (-2A \sin x + 2B \cos x) = \frac{1}{2} \cos x \Rightarrow A = 0, B = \frac{1}{4}$$

$$\therefore y_1^* = \frac{1}{4} x \sin x$$

$$\text{设 } y_2^* = C \cos 3x + D \sin 3x \Rightarrow (-8C \cos 3x - 8D \sin 3x) = -\frac{1}{2} \cos 3x$$

$$\therefore y_2^* = \frac{1}{16} \cos 3x$$

$$\therefore y = C_1 \cos x + C_2 \sin x + \frac{1}{4} x \sin x + \frac{1}{16} \cos 2x$$

$$17. y' = 2x - 1 \quad y'|_{x=0} = -1 = k \Rightarrow y = 1 \quad \therefore \text{切线 } y = -x + 1$$

$$\text{齐次通解: } y = C_1 e^{2x} + C_2 e^x$$

$$2e^x = (2x')e^{1 \cdot x} \quad \lambda = 1 \quad k = 1$$

$$\therefore y^* = X(B)e^x = Bxe^x$$

$$y^{*'} = Be^x + Bxe^x$$

$$y^{*''} = 2Be^x + Bxe^x$$

$$2Be^x + Bxe^x - 3Be^x - 3Bxe^x + 2Bxe^x = 2e^x$$

$$\therefore -B = 2 \Rightarrow B = -2$$

$$\therefore y^* = -2xe^x$$

$$y = C_1 e^{2x} + C_2 e^x - 2xe^x$$

$$y(0) = C_1 + C_2 = 1$$

$$y' = 2C_1 e^{2x} + C_2 e^x - 2e^x - 2xe^x$$

$$y'_{(0)} = 2C_1 + C_2 - 1 = -1$$

$$\therefore 2C_1 + C_2 = 0$$

$$\therefore \begin{cases} C_1 = -1 \\ C_2 = 2 \end{cases} \Rightarrow y = -e^{2x} + 2e^x - 2xe^x$$

$$18. y_1 = 2e^{2x} \quad y_2 = e^x \quad y_3 = xe^x$$

$$\therefore r_1 = 1 \quad r_2 = 2$$

$$\therefore r^2 - 3r + 2 = 0$$

$$\therefore \alpha = -3, \beta = 2 \quad y'' - 3y' + 2y = \gamma e^x$$

$$\therefore y^* = xe^x$$

$$y^{*'} = e^x(x+1)$$

$$y^{*''} = e^x(x+2)$$

$$\therefore e^x(x+2 - 3x - 3) + 2xe^x = \gamma e^x$$

$$-e^x = \gamma e^x \Rightarrow \gamma = -1$$

$$\therefore \begin{cases} \alpha = -3 \\ \beta = 2 \\ \gamma = -1 \end{cases}$$

$$\text{Ans: } y = C_1 e^x + C_2 e^{2x} + xe^x$$

$$20. y'' - (2t+1)y' + t^2 y = 0$$

$$x = \ln t. \quad \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = t \cdot \frac{dy}{dt}$$

$$\therefore \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(t \cdot \frac{dy}{dt} \right) = \frac{dt}{dx} \cdot \frac{dy}{dt} + \frac{d^2 y}{dt^2} \cdot t = t \frac{dy}{dt} + t \frac{d^2 y}{dt^2}$$

$$\therefore y'' = t y_t' + t^2 y_t''$$

$$y' = t y_t'$$

$$\therefore t^3 y_t'' - 2t^2 y_t' + t^2 y = 0$$

$$\therefore \lambda^2 - \lambda + 1 = 0 \Rightarrow y = c_1 e^t + c_2 t e^t$$

$$\therefore y = (c_1 + c_2 e^x) e^{e^x}$$

$$23. \text{ 令 } F(x) = \int_a^x f(t) g(t) dt, \quad F'(x) = f(x) g(x)$$

$$f(x) \leq k F(x) + C \Rightarrow f(x) g(x) = F'(x) \leq k g(x) F(x) + C g(x)$$

$$\therefore F'(x) - k g(x) F(x) \leq C g(x)$$

$$\text{令 } F(x) = y$$

$$\text{有 } y' - k g(x) y \leq C g(x)$$

$$\text{若取等, 有 } y = C(x) e^{\int k g(x) dx} = C(x) e^{k G(x)}$$

$$C(x) = \int g(x) e^{-\int k g(x) dx} dx$$

$$= C \int g(x) e^{-k G(x)} dx$$